

EE 5239 Nonlinear Optimization

Lecture 2: Gradient Methods II: Analysis

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Outline

- 1 Convergence and Rate Analysis: Overview
- 2 Gradient Methods: Analysis

Theoretical Analysis

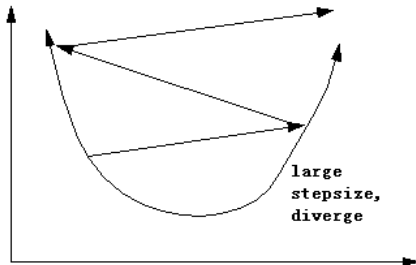
- Analyze convergence behavior, focused on the “Steepest gradient descent” method

$$\mathbf{x}^{r+1} = \mathbf{x}^r - \alpha_r \nabla f(\mathbf{x}^r), \quad r = 0, 1, \dots$$

- **Question:** When does the algorithm converge, to what solution?
- **Question:** How fast does the algorithm converge?

One Missing Piece

- The most commonly used stepsize rule is the **constant rule**
- But how to pick it? (yeah, to achieve descent, but how?)
- If not too carefully....



One Missing Piece

- **Intuition:** stepsize should be *inversely proportional* to the “largest” curvature of the function
[graphically...]

Convergence of Iterative Methods

- **Convergence to stationary solutions**
 - ① Sanity check
 - ② Minimal requirement of any reasonable algorithm
 - ③ Does not give global efficiency of the algorithm
 - ④ Linear rate/Supperlinear rate/Sublinear rate

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- **Iteration complexity analysis (convergence rate)**

- ① Measures the number of iterations required to get an optimal solution (e.g., $f(\mathbf{x}^r) - f(x^*) \leq \epsilon$)
- ② Current analysis is all for the worst case
- ③ Gives global behavior of the algorithm

The Analysis of GD Method

- Suppose exists a constant L , which bounds the maximum eigenvalue of Hessian matrix of f :

$$\nabla^2 f(\mathbf{x}) \preceq L\mathbf{I},$$

where $\mathbf{I}$ is an identity matrix

- This implies that the curvature of the function is bounded
- Then we have

$$\begin{aligned} f(\mathbf{x}) &\stackrel{(i)}{=} f(\mathbf{y}) + \langle \nabla f(\mathbf{y}), \mathbf{x} - \mathbf{y} \rangle + (\mathbf{x} - \mathbf{y})^T \nabla^2 f(\tilde{\mathbf{x}})(\mathbf{x} - \mathbf{y}) \\ &\leq f(\mathbf{y}) + \langle \nabla f(\mathbf{y}), \mathbf{x} - \mathbf{y} \rangle + \frac{L}{2} \|\mathbf{x} - \mathbf{y}\|^2 := u(\mathbf{x}; \mathbf{y}), \quad \forall \mathbf{x}, \mathbf{y} \end{aligned}$$

where the first step is due to **mean value theorem**; the second step is due to the boundedness of Hessian

The Descent Lemma

- The same result, but stated in a slightly different way
- **Key Lemma:** The Descent Lemma

Suppose $f : \mathbb{R}^n \rightarrow \mathbb{R}$ has Lipschitz gradient

$$\|\nabla f(\mathbf{x}) - \nabla f(\mathbf{y})\| \leq L\|\mathbf{x} - \mathbf{y}\|, \forall \mathbf{x}, \mathbf{y} \in \mathbb{R}^n$$

Then we have

$$f(\mathbf{x}) \leq f(\mathbf{y}) + \langle \nabla f(\mathbf{y}), \mathbf{x} - \mathbf{y} \rangle + \frac{L}{2}\|\mathbf{x} - \mathbf{y}\|^2 := u(\mathbf{x}; \mathbf{y}), \forall \mathbf{x}, \mathbf{y}$$

- Read Prop. A. 24 of Bertsekas for proof

Proof of Descent Lemma

- Let t be a scalar and let $g(t) = f(x + ty)$
- Chain rule: $g'(t) = y' \nabla f(x + ty)$
- We have the following

$$\begin{aligned} & f(x + y) - f(x) \\ &= g(1) - g(0) = \int_0^1 g'(t) dt = \int_0^1 y' \nabla f(x + ty) dt \\ &\leq \int_0^1 y' \nabla f(x) dt + \left| \int_0^1 y' (\nabla f(x + yt) - \nabla f(x)) dt \right| \\ &\leq \int_0^1 y' \nabla f(x) dt + \int_0^1 \|y\| \|\nabla f(x + yt) - \nabla f(x)\| dt \\ &\leq y' \nabla f(x) + \|y\| \int_0^1 L t \|y\| dt \quad (\text{Lipschitz continuity}) \\ &= y' \nabla f(x) + \frac{L}{2} \|y\|^2 \end{aligned}$$

Apply the Descent Lemma

- **Example:** for a quadratic problem with $f(\mathbf{x}) = \frac{1}{2}\mathbf{x}^T \mathbf{A}\mathbf{x}$, what is L ? can you verify that the descent lemma is true?
- Replace $\mathbf{y}$ by $\mathbf{x}^r$ in the descent lemma

$$f(\mathbf{x}) \leq f(\mathbf{x}^r) + \langle \nabla f(\mathbf{x}^r), \mathbf{x} - \mathbf{x}^r \rangle + \frac{L}{2} \|\mathbf{x} - \mathbf{x}^r\|^2 := u(\mathbf{x}; \mathbf{x}^r), \quad \forall \mathbf{x}$$

- Minimize the right hand side with respect to $\mathbf{x}$, and let $\mathbf{x}^* = \mathbf{x}^{r+1}$:

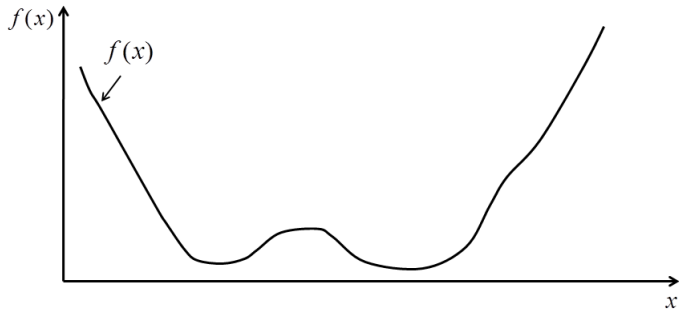
$$\mathbf{x}^{r+1} = \mathbf{x}^r - \frac{1}{L} \nabla f(\mathbf{x}^r)$$

Apply the Descent Lemma

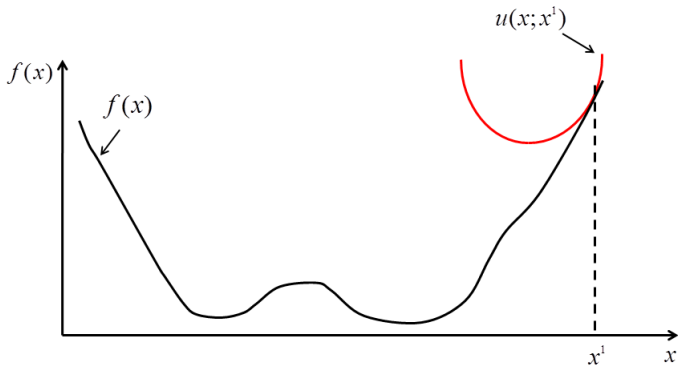
- **Claim** such $\mathbf{x}^{r+1}$ always decreases the objective! Why?
- By how much? Plug the expression $\mathbf{x}^{r+1}$ into the descent estimate:

$$f(\mathbf{x}^{r+1}) \leq f(\mathbf{x}^r) - \frac{1}{2L} \|\nabla f(\mathbf{x}^r)\|^2$$

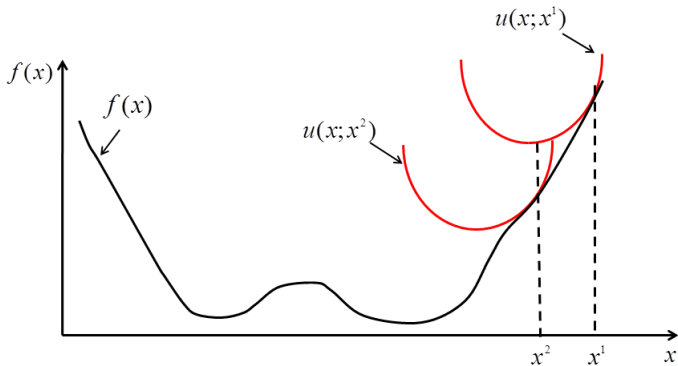
One Missing Piece (illustration)



One Missing Piece (illustration)



One Missing Piece (illustration)



Convergence Analysis

- **Convergence:** Must at least converge to first order optimality ($\nabla f(\mathbf{x}) = 0$)
 - ① Basic requirement
 - ② Not necessarily convergent to global, or local, optimal
 - ③ This only says that the algorithm is reasonable
- Provide an analysis of GD for **constant stepsize rule**, for any reasonable direction (not necessarily $-\nabla f(\mathbf{x})$)

A General Analysis for Convergence

- Prove convergence to points that satisfy the first order optimality
 - we call them **stationary solutions** [figure]
- The direction $\mathbf{d}^r$ cannot be **orthogonal** to $\nabla f(\mathbf{x}^r)$ [figure]

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- The direction $\mathbf{d}^r$ cannot be **orthogonal** to $\nabla f(\mathbf{x}^r)$ [figure]
- **Gradient related condition:** For any sequence $\{\mathbf{x}^r\}$ that converges to a nonstationary point, the corresponding direction $\{\mathbf{d}^r\}$ is bounded and satisfies

$$\lim_{r \rightarrow \infty} \langle \nabla f(\mathbf{x}^r), \mathbf{d}^r \rangle < 0$$

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- Is this condition satisfied for $\mathbf{d}^r = -\mathbf{D}^r \nabla f(\mathbf{x}^r)$ with $\mathbf{D}^r \succ 0$?

A General Analysis of Convergence

- Here we state a few assumptions about the algorithm/problem
- Let $\mathbf{x}^r$ be a sequence generated by a gradient method

$$\mathbf{x}^{r+1} = \mathbf{x}^r + \alpha_r \mathbf{d}^r$$

- $\mathbf{d}^r$ is gradient related
- Assume the following Lipschitz continuous condition is satisfied

$$\|\nabla f(\mathbf{x}) - \nabla f(\mathbf{y})\| \leq L\|\mathbf{x} - \mathbf{y}\|, \forall \mathbf{x}, \mathbf{y} \in \mathbb{R}^n$$

- Note, the problem is not necessarily second-order differentiable

A General Analysis of Convergence (Assumptions)

- Assume **either one** of the following choices of stepsize

- There exists a scalar ϵ such that for all r

$$\epsilon < \alpha_r \leq -\frac{(2 - \epsilon)\langle \nabla f(\mathbf{x}^r), \mathbf{d}^r \rangle}{L\|\mathbf{d}^r\|^2}$$

- $\alpha_r \rightarrow 0$, and $\sum_{r=1}^{\infty} \alpha_r = \infty$ (i.e., $\alpha_r = \frac{1}{r}$)

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- Claim:** $\nabla f(\mathbf{x}^r) \rightarrow 0$, or $f(\mathbf{x}^r) \rightarrow -\infty$
- If $\mathbf{d}^r = -\nabla f(\mathbf{x}^r)$, then the first condition becomes

$$\epsilon < \alpha_r \leq \frac{(2 - \epsilon)}{L}$$

Therefore, we can pick, for example, $\alpha_r = \frac{1}{L}$

Convergence Analysis

- Given $\mathbf{x}^r$ and the descent direction $\mathbf{d}^r$, the Lipschitz assumption implies (cf. the descent Lemma)

$$f(\mathbf{x}^r + \alpha_r \mathbf{d}^r) - f(\mathbf{x}^r) \leq \alpha_r \langle \nabla f(\mathbf{x}^r), \mathbf{d}^r \rangle + \frac{L}{2} \alpha_r^2 \|\mathbf{d}^r\|^2$$

- Plugin the upper-bound for α_r :

$$f(x^r + \alpha_r \mathbf{d}^r) - f(\mathbf{x}^r) \leq \underbrace{-\epsilon(2 - \epsilon)}_{<0} \frac{(\langle \nabla f(\mathbf{x}^r), \mathbf{d}^r \rangle)^2}{2L \|\mathbf{d}^r\|^2}$$

- Clearly, the objective is always **decreasing**
- Question:** Where have we used the “gradient related” condition?

Convergence Analysis (cont.)

- Assume $\bar{x}$ is a finite **nonstationary** limit point.
- Then $f(\mathbf{x}^r) \downarrow f(\bar{\mathbf{x}})$
- Then $\langle \nabla f(\mathbf{x}^r), \mathbf{d}^r \rangle \rightarrow 0$, because otherwise $f(\bar{\mathbf{x}}) \rightarrow -\infty$

Convergence Analysis (cont.)

- Assume $\bar{x}$ is a finite **nonstationary** limit point.
- Then $f(\mathbf{x}^r) \downarrow f(\bar{\mathbf{x}})$
- Then $\langle \nabla f(\mathbf{x}^r), \mathbf{d}^r \rangle \rightarrow 0$, because otherwise $f(\bar{\mathbf{x}}) \rightarrow -\infty$
- Is this possible? No, the gradient related condition asserts that $\langle \nabla f(\mathbf{x}^r), \mathbf{d}^r \rangle < 0$, and the condition on α_r asserts that $\alpha_r > 0$
- We arrived at a contradiction – the claim is proved
- How about the diminishing stepsize? Same analysis, but much more involved

Summary: Convergence Analysis

- **Convergence:** The GD converges to first order optimality ($\nabla f(\mathbf{x}) = 0$)
 - ① Basic requirement
 - ② Not necessarily convergent to globally optimal
 - ③ This only says that the algorithm is **reasonable** (why?)
- Provide an analysis of GD for **constant stepsize rule**, for any reasonable direction (not necessarily $-\nabla f(\mathbf{x})$)

Homework

- Read Section 1.2, especially Section 1.2.2, especially Prop 1.2.1 - 1.2.3 (Will cover the proof of 1.2.1 next time)
- Read Proof Prop. 1.2.4 if you are interested in analysis of **diminishing stepsize** (very delicate proof!)